# Lab 07: Docker Basics and PostgreSQL Containers

CS-649 Cloud DevOps

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| **Course** | CS-649 Cloud DevOps |
| **Section** | A |
| **University** | Fatima Jinnah Women University (FJWU), Rawalpindi |
| **Lab** | Lab 07 - Docker installation, running containers, inspecting and cleanup |
| **GitHub** | github.com/sabagu003-ui/DevOps (Labs/Lab07) |

## Lab Overview

Docker was verified on Ubuntu (WSL), two versions of PostgreSQL were run as containers, Docker information and search commands were used, and a container with its image was restarted, stopped and removed.

## Task 1: Docker Installation

### Verify Docker and Docker Compose

**Objective:** Confirm that Docker Engine and Docker Compose are installed and working.

**Command(s):**

```
docker --version
docker compose version
```

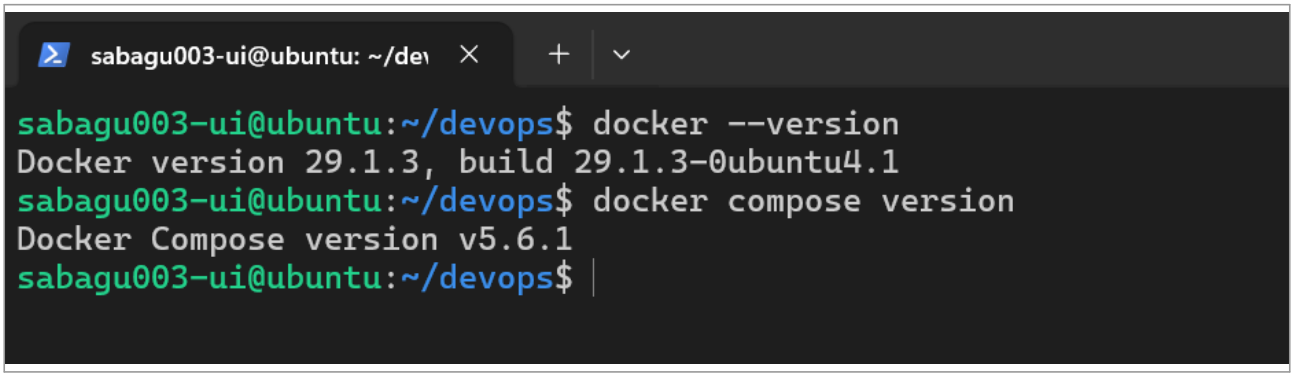



*Figure 1: Docker and Docker Compose versions*

**Explanation:** Docker 29.1.3 and Docker Compose v5.6.1 are installed and working.

# Task 2: Running Two PostgreSQL Versions

## 2.1 Run PostgreSQL 9.6 without a password

**Objective:** Understand what happens when the PostgreSQL container is started without a superuser password.

**Command(s):**

```
docker pull postgres:9.6
docker run postgres:9.6
```

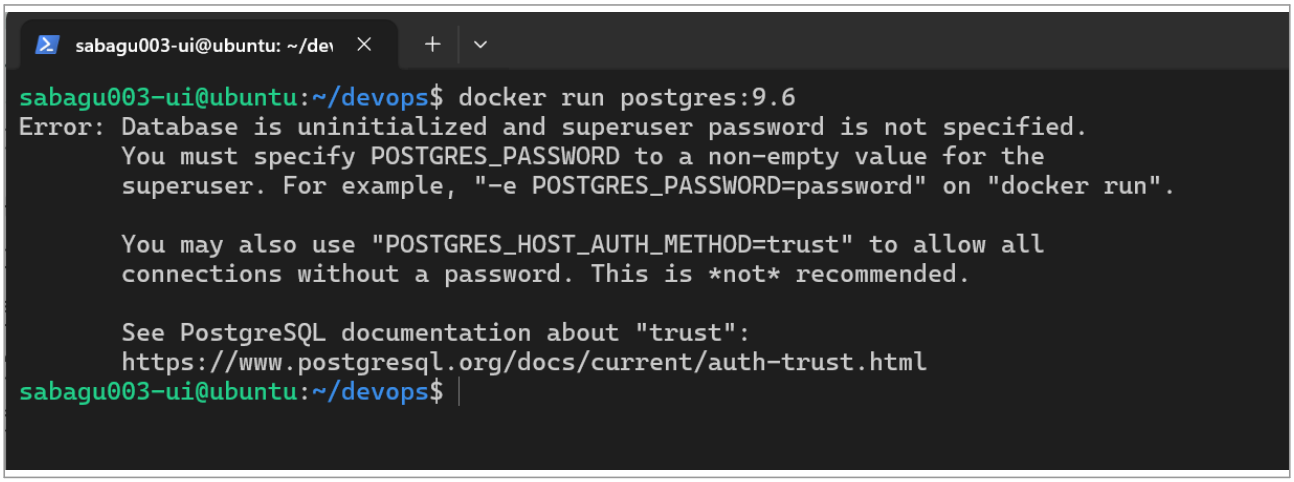



*Figure 2: postgres:9.6 exits with a missing password error*

**Explanation:** The container exits with an error because the PostgreSQL image needs the POSTGRES_PASSWORD variable. This error is expected.

## 2.2 Run PostgreSQL 9.6 with a password

**Objective:** Start PostgreSQL 9.6 in the background with the required environment variable.

**Command(s):**

```
docker run -d -e POSTGRES_PASSWORD=password postgres:9.6
docker ps
```

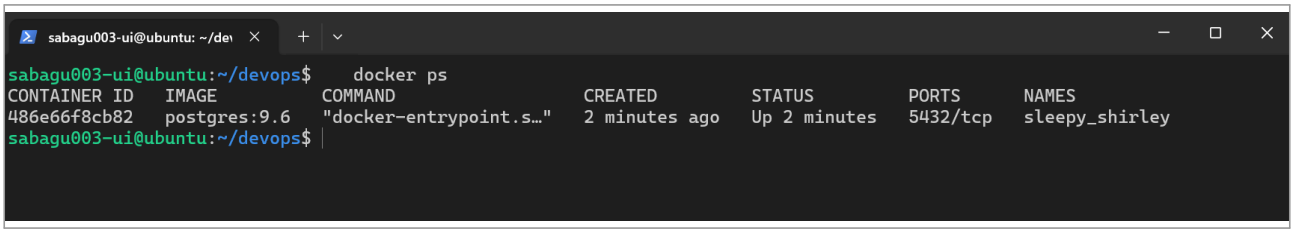



*Figure 3: postgres:9.6 container is Up*

**Explanation:** -d runs the container in the background and -e sets the password. postgres:9.6 shows Up (name: sleepy_shirley).

## 2.3 Run PostgreSQL 10.10 alongside 9.6

**Objective:** Run a second PostgreSQL version at the same time as the first one.

**Command(s):**

```
docker run -d -e POSTGRES_PASSWORD=password postgres:10.10
docker ps
```

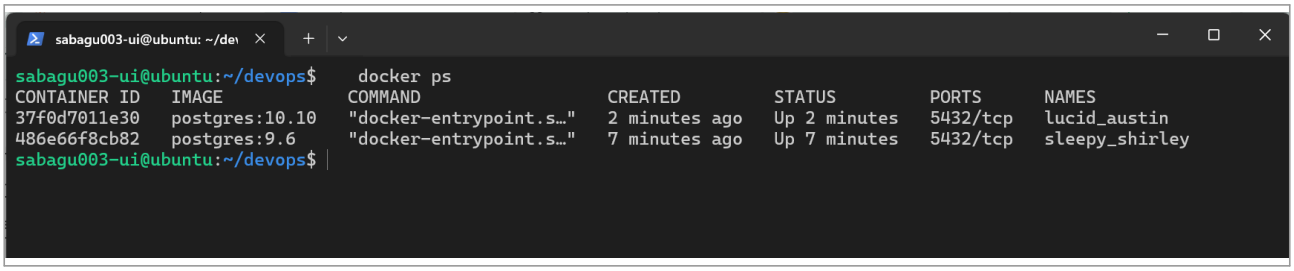



*Figure 4: postgres:10.10 and postgres:9.6 both running*

**Explanation:** postgres:10.10 was downloaded and started. Both versions are Up together as separate containers (lucid_austin and sleepy_shirley).

# Task 3: Docker Inspect and Cleanup

## 3.1 Docker system information

**Objective:** View details of the Docker installation and the host system.

**Command(s):**

```
docker info
```

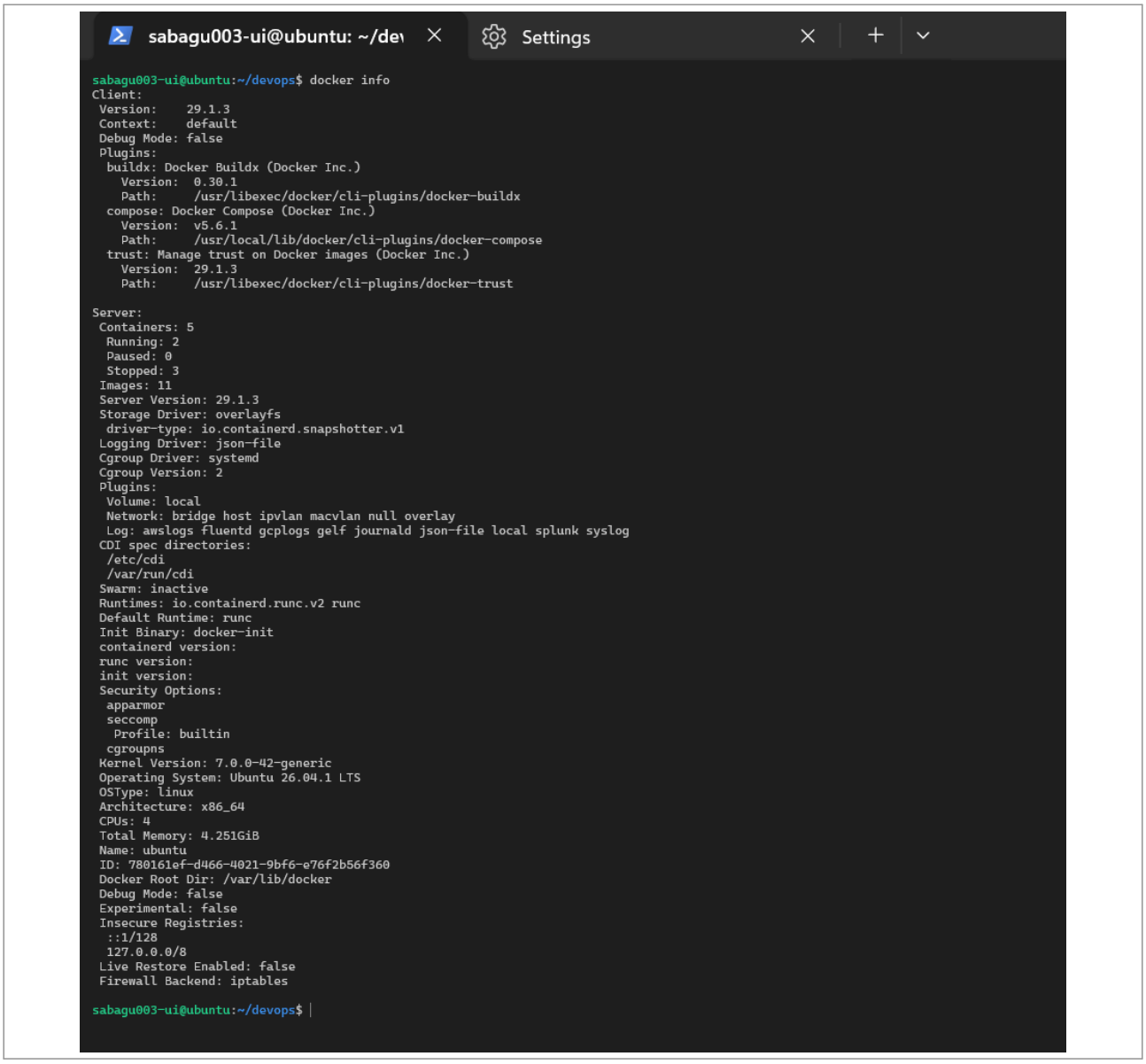



*Figure 5: Output of docker info*

**Explanation:** Server Version 29.1.3, Storage Driver overlayfs, Ubuntu 26.04.1 LTS, 4 CPUs, 4.251 GiB memory. 5 containers (2 running, 3 stopped) and 11 images.

## 3.2 Search Docker Hub for Ubuntu

**Objective:** Search for the Ubuntu image on Docker Hub.

**Command(s):**

```
docker search ubuntu
```

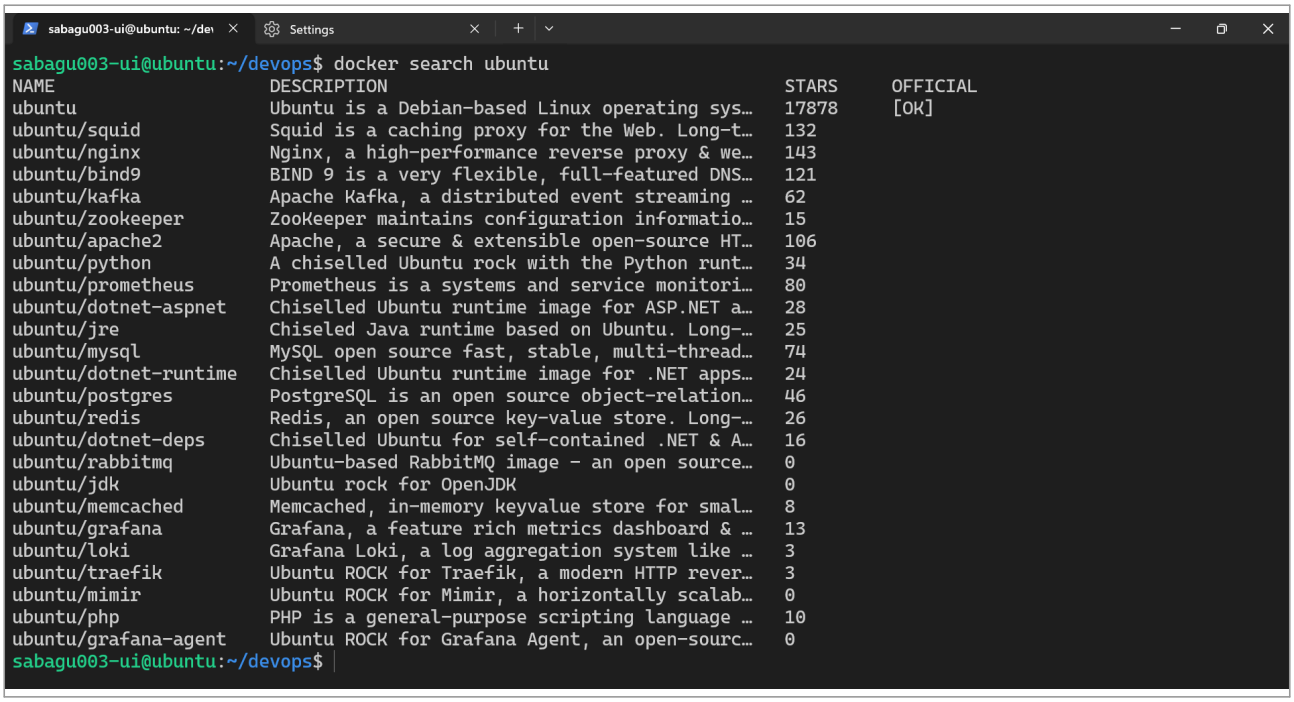



*Figure 6: Output of docker search ubuntu*

**Explanation:** The first row, ubuntu, is the official image ([OK], 17878 stars). The rest are Ubuntu-based images from other publishers.

## 3.3 Ubuntu container status

**Objective:** Run an Ubuntu container in the background and compare docker ps with docker ps -a.

**Command(s):**

```
docker images
docker run -d ubuntu
docker ps
docker ps -a
```

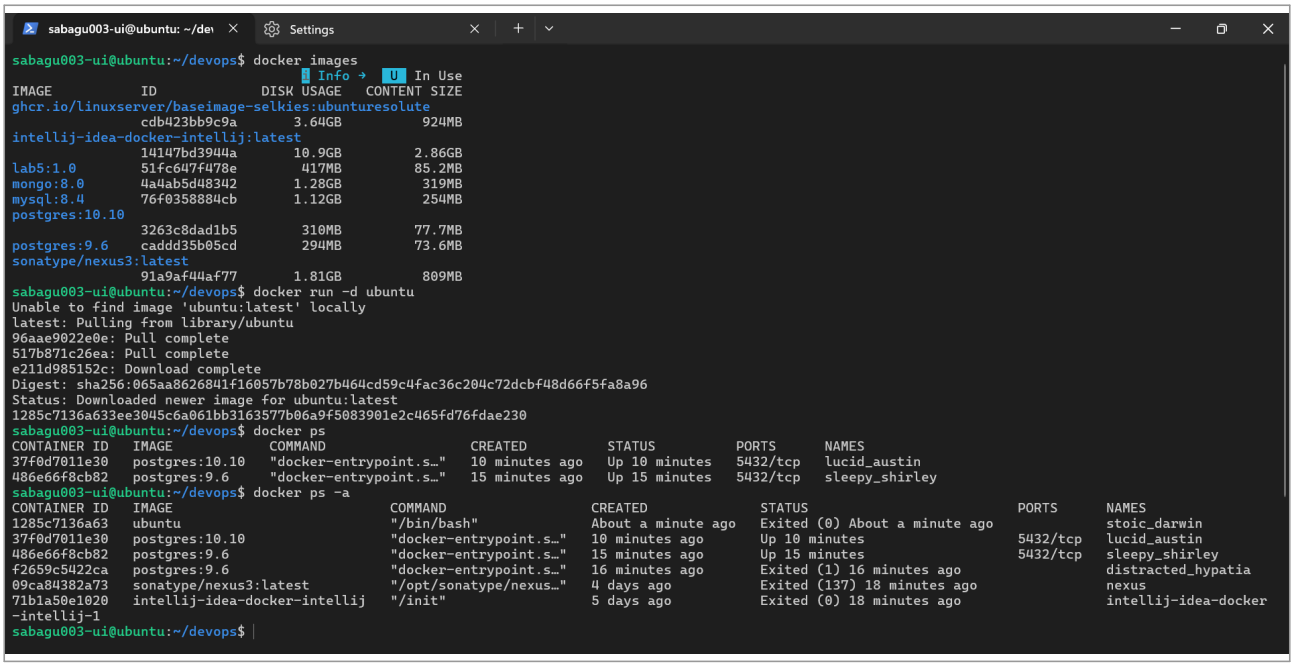



*Figure 7: docker images, docker run -d ubuntu, docker ps and docker ps -a*

**Explanation:** docker run -d ubuntu pulled ubuntu:latest. docker ps does not list it because it already exited. docker ps -a shows it as Exited (0) because /bin/bash has no terminal to keep it running.

### 3.4 Restart, stop and remove a container, then remove its image

**Objective:** Practice the container lifecycle and cleanup for the postgres:9.6 container.

**Command(s):**

```
docker restart sleepy_shirley
docker stop sleepy_shirley
docker rm sleepy_shirley
docker rmi postgres:9.6
```

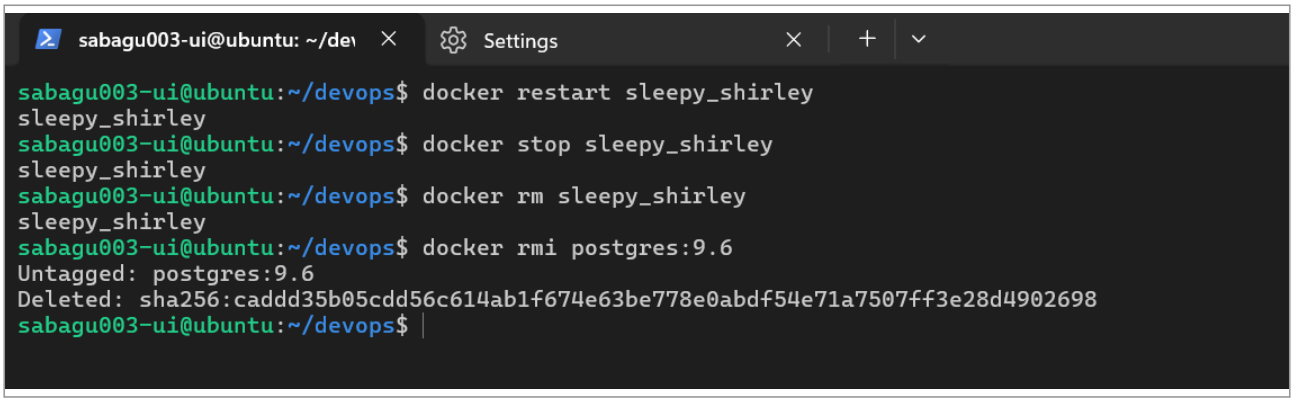



*Figure 8: Cleanup of the postgres:9.6 container and image*

**Explanation:** The container was restarted, stopped and removed; then the image was deleted (Untagged: postgres:9.6). An image can be removed only when no container uses it. postgres:10.10 was left running.

## Conclusion

Docker works correctly, two PostgreSQL versions ran side by side in separate containers, and the container lifecycle (run, restart, stop, remove, rmi) was practiced successfully.